

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 13

Measurement: How the World Gets Numbers

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. “A scale is a homomorphism.” Say precisely from what, to what, and what it preserves. Why does calling it a homomorphism connect measurement to everything in Phase III?

2. Two analysts compute a statistic on the same ordinal data; one rescales the categories first (monotonically). They reach opposite conclusions. Which analyst, if either, is wrong, and how does the invariance criterion decide?

3. Why is “meaningfulness” in measurement the *same idea* as “invariance” in Chapter 5? State the shared structure in one sentence.

4. A colleague insists you must *never* average ordinal data. Using Velleman & Wilkinson, give the strongest counter-case, and then state the condition under which they are nonetheless right.

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
